



PRO Cook: Personalized Recipe Organizer

In Partial Fulfillment of the Requirements in
CST14/L – Fundamentals of Cybersecurity

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CHAPTER 1: INTRODUCTION

Overview of the System

This study focuses on the development and evaluation of PRO Cook (Personalized Recipe Organizer), a system designed to help users store, organize, search, and explore personal recipe collections efficiently. The system provides a user-friendly interface that allows users to manage recipes and sort them dynamically based on preparation time and number of ingredients. By organizing recipe data in a structured manner, PRO Cook aims to improve accessibility and ease of use compared to traditional recipe management methods.

The study implements an array-based data structure and utilizes the insertion sort algorithm to perform sorting operations within the system. To evaluate the effectiveness of the algorithm, simulation tests are conducted to measure execution time under different sorting conditions, including ascending and descending order. The results of the evaluation are analyzed to determine the suitability of insertion sort for managing recipe data. Through this approach, the study highlights the practical application of data structures and algorithms while addressing common challenges in personalized recipe organization.

Background of the Study

With the growing reliance on digital applications for managing personal information, individuals increasingly seek convenient and efficient ways to organize their daily activities, including cooking and meal preparation. Many cooking enthusiasts still rely on handwritten notes, printed recipes, or unstructured digital files to store their recipes. These traditional methods often result in difficulties in organizing, searching, and retrieving recipe information, especially when users want to access recipes based on specific criteria such as preparation time, number of ingredients, or meal type.

In the field of computer science, data structures and algorithms play a crucial role in improving the efficiency and performance of information systems. Sorting algorithms, in particular, are essential for organizing data in a way that enhances accessibility and usability. Insertion sort is a simple and reliable algorithm that performs efficiently on small to medium-sized datasets and supports in-place sorting without additional memory requirements. Since personal recipe collections typically involve manageable amounts of data, insertion sort is suitable for implementation in a personalized recipe management system. This study integrates these concepts into the development of a practical application to demonstrate the real-world use of data structures and algorithms.

Statement of the Problem

Managing personal recipe collections using traditional methods or unorganized digital files often makes it difficult for users to store, search, and retrieve recipes efficiently. Many users need to organize recipes based on criteria such as preparation time, number of ingredients, and meal type; however, existing recipe management solutions often lack effective and flexible sorting features. As a result, accessing relevant recipes becomes time-consuming and less convenient for cooking enthusiasts.

Moreover, the practical application of data structures and algorithms in simple information systems is not always clearly demonstrated. There is a need to implement and evaluate appropriate data structures and sorting algorithms that can efficiently handle recipe data. In particular, the insertion sort algorithm must be examined to determine its suitability for sorting small to medium-sized datasets based on different criteria and orderings. This study addresses these concerns by developing the PRO Cook (Personalized Recipe Organizer) system and analyzing the performance of insertion sort in organizing recipe information.

General Objectives

To develop PRO Cook (Personalized Recipe Organizer), a user-friendly and efficient platform that enables cooking enthusiasts to store, organize, share, and explore recipes, making recipe management simple and collaborative.

Specific Objectives

1. To allow users to save and manage their personal recipe collections in a centralized platform.
2. To provide features that enable users to share recipes and discover new dishes from other users.
3. To design an intuitive and easy-to-use interface for seamless recipe navigation.
4. To support searching, sorting, and filtering of recipes based on ingredients, cooking time, and difficulty level.
5. To improve the overall efficiency and convenience of managing recipes compared to traditional methods.

CHAPTER 2: REVIEW OF RELATED LITERATURE AND SYSTEMS

Related Studies and Systems

2.1 Mobile Recipe and Nutrition Applications

The development of mobile applications has significantly changed how individuals manage food preparation and dietary habits. Many users now rely on smartphone applications to search for recipes, track ingredients, and monitor their eating behaviors. Research shows that mobile-based food and nutrition applications positively influence user engagement and dietary decisions.

According to *Seid et al. (2024)*, smartphone-based nutrition applications improve healthy eating behaviors and overall diet quality. Their systematic review of randomized controlled trials found that users who utilized mobile nutrition applications demonstrated better food choices compared to those who did not use such tools. This finding suggests that structured digital systems can effectively support users in organizing and managing food-related information.

Similarly, *Nogueira-Rio et al. (2024)* emphasized that integrating artificial intelligence (AI) into mobile nutrition applications enhances personalization and user engagement. Their study explained that AI allows applications to provide tailored recommendations based on user preferences and dietary needs. These findings support the idea that personalized systems, such as PRO Cook, can improve the user experience by offering customized recipe suggestions and filtering options.

Furthermore, *Bondevik (2024)* conducted a systematic review of food recommender systems and highlighted that modern culinary platforms increasingly use algorithm-based recommendation models. These systems analyze user data and preferences to suggest relevant recipes. This study provides theoretical support for incorporating sorting and filtering mechanisms into recipe organizer systems.

2.2 Personalized Recommendation Systems in Culinary Platforms

Personalization is a key feature in modern digital platforms. Recipe applications that adapt to user preferences tend to improve usability and satisfaction. Machine learning and natural language processing are commonly used to enhance personalization features.

Xing and Gao (2024) introduced RecipeRadar, an AI-powered recipe recommendation system that interprets user input and suggests relevant recipes based on available ingredients. Their research demonstrates how structured data and intelligent ranking algorithms can improve recipe search results. This concept aligns with the objective of PRO Cook, which aims to provide efficient sorting and filtering of stored recipes.

In addition, *Tsolakidis et al. (2024)* reviewed artificial intelligence and machine learning technologies used in personalized nutrition systems. The authors explained that analyzing user behavior and dietary patterns allows systems to generate more accurate recommendations. This supports the importance of implementing organized data structures and efficient sorting methods in recipe management systems.

2.3 Data Structures and Sorting Algorithms in Information Systems

Efficient data organization is essential for information systems. Sorting algorithms help arrange data in a structured manner, making retrieval faster and more convenient. According to *Cormen et al. (2009)*, sorting plays a central role in improving system performance. The authors explain that algorithm selection should depend on dataset size and system requirements.

Insertion sort is particularly suitable for small or partially sorted datasets. *Sedgewick and Wayne (2011)* state that insertion sort performs efficiently for small-scale data and requires minimal memory usage. Since personal recipe collections typically contain a manageable number of entries, insertion sort is appropriate for implementation in PRO Cook.

Knuth (1998) also emphasized the importance of evaluating algorithm efficiency through performance testing. Measuring execution time under different sorting conditions ensures that the algorithm performs effectively in real-world applications. This supports the study's approach in testing insertion sort within the system.

2.4 User Interface Design and Usability in Food Applications

User-centered design is important in developing applications that are easy to use and understand. *Norman (2013)* explained that systems should be intuitive and reduce cognitive effort for users. Clear organization, logical categorization, and responsive filtering improve overall usability.

Ngelambong et al. (2024) analyzed negative user experiences in food and beverage mobile applications. Their findings showed that poor navigation and complex filtering options can reduce user satisfaction. This highlights the importance of designing simple and efficient interfaces in recipe organizer systems.

Shneiderman et al. (2016) further explained that interactive systems should provide flexible searching, sorting, and filtering options. These features empower users to control how they access and manage their data. The PRO Cook system applies these principles by allowing users to sort recipes based on preparation time, ingredients, and difficulty level.

2.5 Security Considerations in Recipe Management Systems

Although recipe organizers may seem simple, they still require basic security measures. *Stallings (2018)* explained that information systems must protect confidentiality, integrity, and availability of data. Even small applications must implement authentication and secure data storage to prevent unauthorized access.

The Open Web Application Security Project (OWASP, 2021) identified common security risks in web-based applications, such as injection attacks and broken authentication. Following secure coding practices helps protect user accounts and stored recipe information in systems like PRO Cook.

Security Issues in Existing Systems

While mobile recipe and nutrition applications provide convenience and personalization, existing systems are not free from security risks. As discussed in the previous sections, many modern food and recipe platforms rely on user accounts, stored preferences, and cloud-based databases. These features, although beneficial for personalization, also introduce cybersecurity vulnerabilities.

According to *Stallings (2018)*, information systems must protect three fundamental principles: confidentiality, integrity, and availability. In the context of recipe and nutrition applications, confidentiality refers to protecting user account credentials and personal dietary information. Integrity ensures that stored recipes and user data are not altered without authorization. Availability guarantees that users can access their stored recipes when needed. Failure to implement proper security mechanisms may compromise these principles.

The *Open Web Application Security Project (OWASP, 2021)* identified the most common vulnerabilities affecting web and mobile applications. These include broken authentication, injection attacks, insecure design, and insufficient logging and monitoring. Many small-scale applications, particularly student-developed or independent systems, often overlook secure coding practices such as input validation and password hashing. This creates potential risks of unauthorized access or data manipulation.

Research also shows that mobile applications frequently experience data privacy and security concerns. According to *Tsolakidis et al. (2024)*, AI-driven personalized nutrition systems collect and analyze user behavior data to generate recommendations. While this improves personalization, it also increases the amount of sensitive data stored within the system. Without proper encryption and access control, such systems may expose users to privacy breaches.

Similarly, *Seid et al. (2024)* emphasized that nutrition applications store health-related data, which may be considered sensitive personal information. Applications that fail to implement adequate data protection measures may violate privacy standards or expose users to risks such as data leakage.

In addition, *Bondevik (2024)* explained that food recommender systems depend heavily on structured datasets and user profiling. The collection of behavioral and preference data introduces potential vulnerabilities if proper security frameworks are not integrated during system development. Recommendation systems must therefore balance personalization with privacy protection.

From a usability perspective, *Ngelambong et al. (2024)* found that some food and beverage mobile applications fail to clearly communicate how user data is stored

and managed. Poor transparency in privacy policies can reduce user trust and raise security concerns.

These findings demonstrate that existing recipe and food-related systems face common cybersecurity challenges, including:

- Weak authentication mechanisms
- Lack of encryption for stored data
- Insufficient input validation
- Poor access control implementation
- Limited transparency in data handling

Although many commercial platforms implement advanced security measures, smaller or newly developed systems may not prioritize cybersecurity during initial development stages.

In the case of PRO Cook, recognizing these security issues is essential. By incorporating secure authentication, input validation, structured database design, and adherence to OWASP guidelines, the system can minimize vulnerabilities while maintaining usability and personalization features. Integrating cybersecurity principles during development ensures that PRO Cook does not replicate the common weaknesses observed in existing systems.

CHAPTER 3: SYSTEM ANALYSIS AND REQUIREMENTS

3.1 Existing System Analysis

Most existing recipe and food management systems — such as mobile nutrition apps, culinary recommendation platforms, and online cooking portals — allow users to browse, search, and save recipes for later use. These systems typically collect user preferences (e.g., favorite cuisines, dietary restrictions), interaction data (e.g., viewed or saved recipes), and sometimes health-related metrics (e.g., caloric targets or nutritional goals). Many of these platforms also use recommendation features that adapt content based on past behavior, enabling personalized suggestions (*Bondevik, 2024; Tsolakidis et al., 2024*).

However, these applications often operate as centralized web or mobile systems with complex backend infrastructures. They rely on cloud databases to store user accounts and content, and usually integrate machine learning or algorithmic filtering to tailor suggestions (*Xing & Gao, 2024; Nogueira-Rio et al., 2024*). Although these features create convenient and engaging experiences, they also introduce complexity in how user data is stored, processed, and secured.

Existing systems vary in their user interface design and functionality, but common capabilities include:

- Recipe search based on keywords or ingredients
- Filtering recipes by category (e.g., dietary preference, cooking time)
- Recommendations based on user history
- Some form of account login and data persistence

Despite these functionalities, reviews and research indicate a number of recurring usability and security problems.

3.2 Problems Identified in the Existing System

By analyzing current literature and real-world systems, several system issues have been consistently documented. These can be categorized as security-related and other technical/usability issues.

3.2.1 Security-Related Issues

Chapter 2 showed that many existing mobile and web-based recipe systems do not adequately address cybersecurity concerns. These issues have been reported in both academic and industry sources as common vulnerabilities in mobile applications.

a. Inadequate Authentication and Authorization

Many applications lack robust authentication controls, which can expose user accounts to unauthorized access. According to the Open Web Application Security Project (OWASP, 2021), *broken authentication* remains a top risk in web and mobile applications. This means users with weak password policies, missing multi-factor authentication (MFA), or unsecured login APIs are more vulnerable to account compromise.

b. Poor Input Validation and Data Sanitization

Applications that accept user input (such as recipe titles, notes, search queries) without proper validation risk *injection attacks* and database corruption. OWASP (2021) emphasizes that improper validation can allow attackers to manipulate backend systems, which may lead to unauthorized data exposure.

c. Insufficient Encryption and Data Protection

Systems that store user credentials or personal preferences without proper encryption may expose sensitive data in the case of a breach. Stallings (2018) explains that encryption of stored data and communications is a fundamental requirement for protecting confidentiality. Many small or emerging applications do not implement strong encryption for stored data at rest or during transmission.

d. Lack of Clear Privacy Policies

Ngelambong et al. (2024) found that several food and beverage applications do not clearly communicate how user data is collected, used, or protected. Poor transparency can lead users to unknowingly share sensitive information, undermining trust and violating data protection expectations.

These security issues illustrate that even well-designed systems risk exploitation when cybersecurity is not prioritized during development

3.2.2 Other Issues (Not Security Related)

Beyond security concerns, existing recipe and food applications exhibit other limitations in usability, performance, and functionality.

a. Limited Personalization Accuracy

Although current systems use recommendation models, the accuracy and relevance of suggested recipes can vary significantly. Bondevik (2024) notes that food recommender systems still struggle with cold-start problems (limited data for new users) and context-aware recommendations (e.g., accounting for meal timing or dietary restrictions).

b. Poor Sorting and Filtering Options

Many applications lack flexible sorting and filtering capabilities. Users often cannot sort recipes by specific criteria such as preparation time, ingredient availability, or difficulty level. This limits usability, as users must manually browse large collections to find relevant content (Shneiderman et al., 2016).

c. Suboptimal User Interface Design

Norman (2013) and Shneiderman et al. (2016) emphasize that intuitive interfaces improve user satisfaction. However, several existing recipe apps have complex menus or inefficient navigation, making it difficult for users to quickly access desired features.

d. High Dependency on External Content

Many current systems depend on third-party content sources (e.g., public recipe databases or web scraped content). This reliance can lead to inconsistent user experience, inconsistency in dietary metadata, and a lack of customization for personal cooking preferences.

e. Lack of Offline Access

Most existing systems require an active internet connection to function, preventing users from accessing their saved recipes when offline. This contrasts with personal recipe organizers that prioritize local data storage and quick access.

3.3. Functional Requirements

- The system shall allow Customers, Business Owners, and Riders to register an account.
- The system shall allow registered users to log in and log out securely.
- The system shall validate login credentials before granting access.
- The system shall display different dashboards based on user roles (Customer, Business Owner, Rider).
- The system shall restrict access to pages according to user role.
- The system shall allow customers to browse available food items.
- The system shall allow customers to select food and place orders.
- The system shall allow customers to track order status.
- The system shall allow customers to view their order history.
- The system shall allow business owners to add, update, and delete food items.
- The system shall allow business owners to view incoming orders.
- The system shall allow business owners to update order status (e.g., preparing, ready).
- The system shall allow riders to view assigned delivery orders.
- The system shall allow riders to update delivery status.
- The system shall allow riders to view delivery history and earnings.
- The system shall notify users when order status changes.

3.4 Non-Functional Requirements

- The system shall use password hashing for storing user credentials.
- The system shall use prepared statements to prevent SQL injection.
- The system shall validate and sanitize all user inputs.
- The system shall implement role-based access control.
- The system shall manage user sessions securely.
- The system shall prevent unauthorized access to restricted pages.
- The system shall respond to user requests within an acceptable time (e.g., less than 3 seconds under normal load).
- The system shall handle multiple users simultaneously.
- The system shall provide a user-friendly interface.
- The system shall ensure dashboards are clearly structured and easy to navigate.
- The system shall maintain data consistency in the database.
- The system shall handle system errors without crashing.
- The system shall use structured and modular PHP code.
- The system shall be accessible through a web browser.
- The system shall ensure minimal downtime.

3.5 User Roles and Access Levels

The system consists of three main user roles:

3.5.1 Customer

Access Level:

- Register and log in
- Browse food items
- Place orders
- Track order status
- View personal order history
- Log out

Restrictions:

- Cannot access business owner dashboard
- Cannot manage food items
- Cannot view other customers' data

3.5.2 Business Owner

Access Level:

- Register and log in
- Manage food menu (add, edit, delete items)
- View customer orders
- Update order status
- Log out

Restrictions:

- Cannot access rider dashboard
- Cannot view other business owners' data
- Cannot access customer personal information beyond order details

3.5.3 Rider

Access Level:

- Register and log in
- View assigned delivery orders
- Update delivery status
- View delivery history and earnings
- Log out

Restrictions:

- Cannot manage food menu
- Cannot access business owner dashboard
- Cannot access customer personal data beyond delivery details

Updated	GUI	PROCOOK	Link:
https://docs.google.com/document/d/1IVCd_eqdDWTlw6jyf0wMy0JcmFHDFy0M_Ekkm3onwjE/edit?usp=drivesdk			

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